

DESCRIPTION

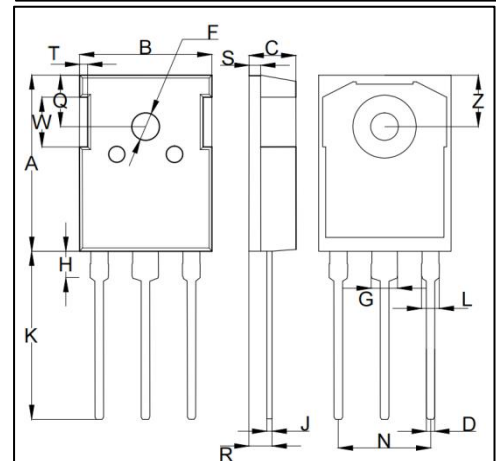
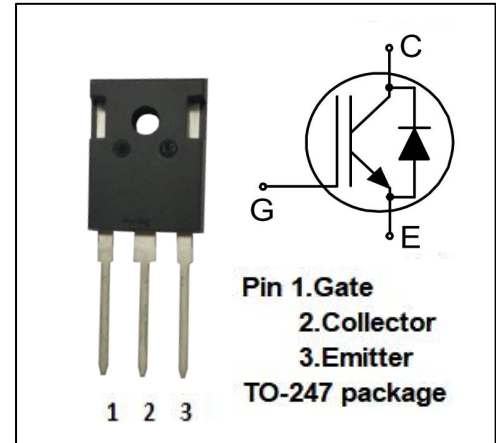
- Low Saturation Voltage: $V_{CE(sat)}=2.56V@I_C=20A$
- High Current Capability
- High Input Impedance
- Fast Switching

APPLICATIONS

- Induction Heating, UPS
- AC & DC motor controls and general purpose inverters

ABSOLUTE MAXIMUM RATINGS

SYMBOL	PARAMETER	VALUE	UNIT
V_{CES}	Collector-Emitter Voltage	1200	V
V_{GES}	Gate-Emitter Voltage	± 20	V
I_C	Collector Current-Continuous @ $T_C=25^\circ C$	45	A
I_C	Collector Current-Continuous @ $T_C=100^\circ C$	24	A
I_{CM}	Pulsed Collector Current @ $T_C=25^\circ C$	180	A
P_D	Power Dissipation , $T_C=25^\circ C$	200	W
T_j	Max. Operating Junction Temperature	-55~150	$^\circ C$
T_{stg}	Storage Temperature Range	-55~150	$^\circ C$



DIM	mm		
	MIN	TYP.	MAX
A	19.80	20.65	21.50
B	15.40	15.65	15.90
C	4.70	5.00	5.30
D	0.90	1.08	1.26
F	3.50	3.70	3.90
G	2.70	3.00	3.30
H	3.90	4.00	4.10
J	0.50	0.60	0.70
K	19.50	20.00	20.50
L	1.90	2.05	2.20
N	10.80	10.90	11.00
Q	6.00	6.15	6.30
R	2.90	3.10	3.30
S	1.80	2.00	2.20
T	2.15	2.25	2.35
W	4.90	5.00	5.10
Z	6.00	6.15	6.30

THERMAL CHARACTERISTICS

SYMBOL	PARAMETER	MAX	UNIT
R _{th j-c}	Thermal Resistance, Junction to Case IGBT	0.64	°C/W
R _{th j-c}	Thermal Resistance, Junction to Case Diode	0.83	°C/W

ELECTRICAL CHARACTERISTICS

SYMBOL	PARAMETER	CONDITIONS	MIN	TYPE	MAX	UNIT
V _{CES}	Collector-Emitter Breakdown Voltage	V _{GE} =0; I _C = 0.25mA	1200	--	--	V
V _{GE(TH)}	Gate-Emitter Threshold Voltage	V _{GE} = V _{CE} ; I _C =0.25mA	3.0	--	6.0	V
V _{CE(sat)}	Collector-Emitter Saturation Voltage	I _C = 20A; V _{CE} = 15V, T _J = 25°C	--	2.56	3.5	V
V _{CE(sat)}	Collector-Emitter Saturation Voltage	I _C = 24A; V _{CE} = 15V, T _J = 25°C	--	2.78	3.7	V
V _{CE(sat)}	Collector-Emitter Saturation Voltage	I _C = 45A; V _{CE} = 15V, T _J = 25°C	--	3.20	--	V
V _{CE(sat)}	Collector-Emitter Saturation Voltage	I _C = 45A; V _{CE} = 15V, T _J = 150°C	--	5.54	--	V
I _{CES}	Zero Gate Voltage Collector Current	V _{CE} = 1200V; V _{GE} =0	--	--	250	uA
I _{CES}	Zero Gate Voltage Collector Current	V _{CE} = 1200V; V _{GE} =0 T _J =150°C	--	--	6500	uA
I _{GES}	Gate-Emitter Leakage Current	V _{GE} = ±20V; V _{CE} =0	--	--	±100	nA
V _{FM}	Diode Forward Voltage Drop	I _C = 16A	--	2.5	3.5	V
V _{FM}	Diode Forward Voltage Drop	I _C = 16A T _J =150°C	--	2.1	3.0	V
C _{ies}	Input Capacitance	V _{GS} = 0V, V _{CS} = 30V, f = 1.0MHz	--	4100	--	pF
C _{oes}	Output Capacitance		--	240	--	
C _{res}	Reverse Transfer Capacitance		--	48	--	
Q _g	Total Gate Charge	V _{GE} = 15V, I _C = 27A, V _{CE} = 400V	--	180	--	nC
Q _{gs}	Gate-Source Charge		--	29	--	
Q _{gd}	Gate-Drain Charge		--	62	--	

SYMBOL	PARAMETER	CONDITIONS	MIN	TYPE	MAX	UNIT
$t_{d(on)}$	Turn-on Delay Time	$V_{GE} = 15V,$ $I_C = 24A,$ $V_{CC} = 800V,$ $R_G = 5.0\Omega$ $T_J = 25^\circ C$	--	48	--	ns
t_r	Turn-on Rise Time		--	28	--	ns
$t_{d(off)}$	Turn-off Delay Time		--	120	170	ns
t_f	Turn-off Fall Time		--	190	260	ns
E_{on}	Turn-on switching losses		--	2.3	--	mJ
E_{off}	Turn-off switching losses		--	1.9	--	mJ
E_{ts}	Total switching losses		--	4.1	--	mJ
$t_{d(on)}$	Turn-on Delay Time	$V_{GE} = 15V,$ $I_C = 24A,$ $V_{CC} = 800V,$ $R_G = 5.0\Omega$ $T_J = 150^\circ C$	--	49	--	ns
t_r	Turn-on Rise Time		--	29	--	ns
$t_{d(off)}$	Turn-off Delay Time		--	245	--	ns
t_f	Turn-off Fall Time		--	340	--	ns
t_{rr}	Reverse Recovery Time	$V_R = 200V$ $di/dt = 200 A/\mu s,$ $I_F = 16A$	--	90	135	ns
Q_{rr}	Reverse Recovery Charge		--	260	675	nC
I_{rrm}	Peak Reverse Recovery Current		--	5.8	10	A

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